Causal statistics for treatment models

3. Difference-in-differences

Ecole Normale Supérieure–PSL, L3 Economics and Cogmaster, 2025

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Today

We will explore the use of difference-in-differences to estimate causal effects. At the end of this lecture, you should:

- Have a good understanding of the difference-in-differences estimation strategy;
- Know the assumptions associated with the simple DiD setting;
- Be able to apply a simple DiD and to program it in R;
- Know about cases when you should be more careful in the use of DiD.

Working with observational data

Last session: RCTs

RCTs = experimental setting. Identification of causal effects by design.

Sometimes, impossible to use an experiment to estimate a treatment effect (unethical, insufficient political willingness, policy evaluation of a program already running).

$\Rightarrow$ Can only use **observational data**. We will explore a method to estimate a treatment effect (the ATT) in this context.

Do you recall what the ATT is?

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Do you recall what the ATT is?

The average treatment effect on the treated (sometimes called TOT):
$ATT = \mathbb{E}\left(Y(1) - Y(0) \mid D = 1\right)$.

Suppose we are using observational data. But we have extremely good reasons to think that the treatment dummy is as good as randomly assigned. What can we do?

As-good-as random assignment

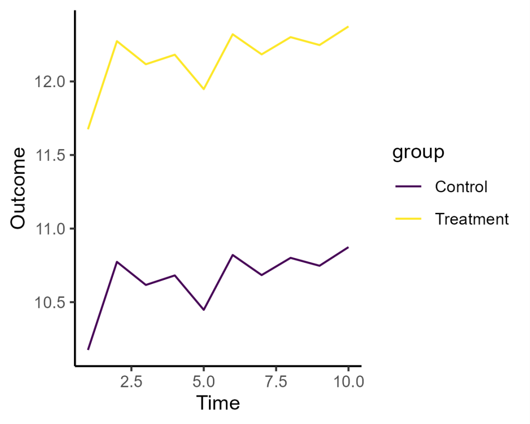
If the treatment is randomly assigned (even if it's not by the design of a RCT implemented by a research team), we can proceed with the same analysis as with an RCT.

However, **As-good-as random assignment is an extremely strong claim**, that should not be made unless there are excellent reasons to think this assumption is true.

Can you think of a weaker assumption than as-good-as random tratment assignment that we could use to estimate a treatment effect?

Parallel trends

Maybe we can find a group that would have evolved similarly as the treatment group, had the treatment group been untreated.



Introducing the difference-in-differences (DiD or diff-in-diff) estimator

The difference-in-differences estimator: an idea

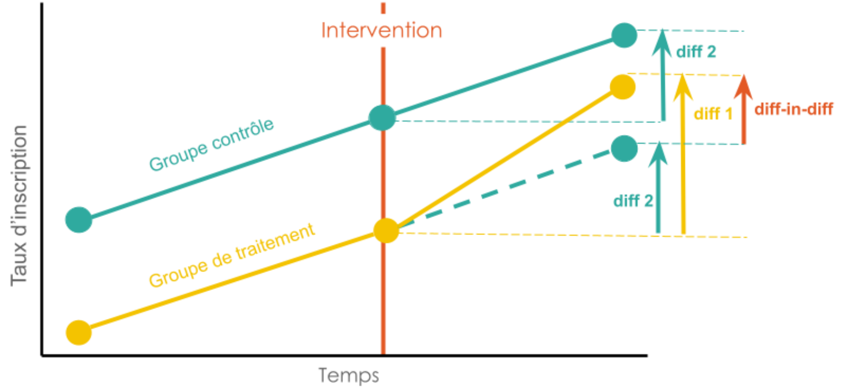


Figure 1: Graphical explanation of the DiD estimator (credit: IDEE program)

Example: a foundational paper

Card, David, and Alan B. Krueger. 1994. "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania." The American Economic Review 84 (4): 772–93.

Context:

- In 1992, New Jersey increased its minimum wage from \$4.25 to \$5.05 per hour.
- Pennsylvania, a neighboring state, did not change its minimum wage.
- Research Question: What is the impact of increasing the minimum wage on employment in the fast-food industry?

Design of the study:

- **Treatment Group:** Fast-food restaurants in New Jersey.
- **Control Group:** Fast-food restaurants in Pennsylvania.
- **Assumption:** Absent the minimum wage increase, employment trends in NJ and PA would have been parallel.

Example: a foundational paper

The DiD estimator:

$$\text{DiD} = \Delta Y_{NJ} - \Delta Y_{PA}$$

where:

- ΔY_{NJ} = Change in employment in NJ.
- ΔY_{PA} = Change in employment in PA.

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Table 1: Mean Full-Time Equivalent (FTE) Employment

State	Pre-Reform Mean FTE	Post-Reform Mean FTE
New Jersey (Treatment)	20.44	21.03
Pennsylvania (Control)	23.33	21.17

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What is the DiD estimate?

Example: a foundational paper

The DiD estimator:

$$\text{DiD} = \Delta Y_{NJ} - \Delta Y_{PA} = 2.75$$

where:

- ΔY_{NJ} = Change in employment in NJ = 0.47.
- ΔY_{PA} = Change in employment in PA = -2.28.

Example: a foundational paper

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Finding: Increase in minimum wage did not decrease employment, and might have improved it.

Warning: paper is old and did not show trends prior to the reform. In a reply to critics, Card and Krueger (2000) showed trends, and used better data (which do support no effect on employment, but are inconsistent with a positive effect of the reform).

Example: a foundational paper

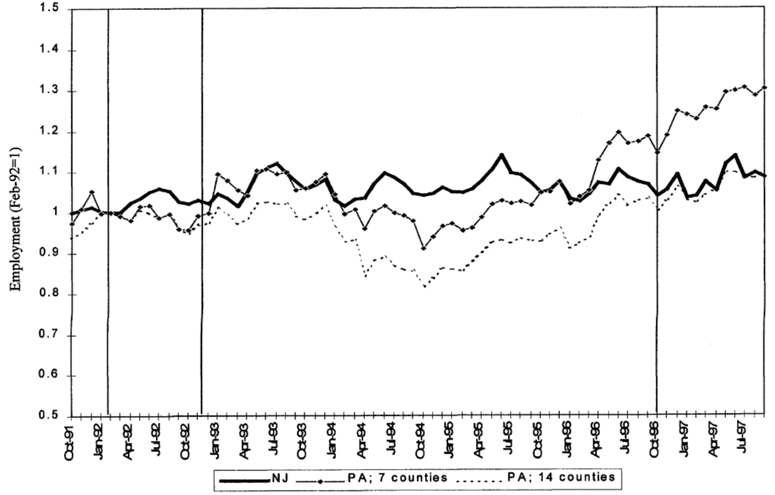

FIGURE 2. EMPLOYMENT IN NEW JERSEY AND PENNSYLVANIA FAST-FOOD RESTAURANTS, OCTOBER 1991 TO SEPTEMBER 1997

Note: Vertical lines indicate dates of original Card-Krueger survey and the October 1996 federal minimum-wage increase.
Source: Authors' calculations based on BLS ES-202 data.

Questions?

Some formalization

The DiD estimator: Notations

- Assume $Y_{i,t}(d)$ is the potential outcome for individual i at time t, with treatment status d.
- For the sake of simplicity, we assume $t \in \{0,1\}$, $d \in \{0,1\}$. I will relax this assumption at the end of the lecture.

The parallel trends assumption

The parallel trends assumption is about what would happen in the absence of treatment.

Parallel trends assumption

$$\mathbb{E}\left(Y_{t=1}(0) - Y_{t=0}(0) \mid D = 1\right) = \mathbb{E}\left(Y_{t=1}(0) - Y_{t=0}(0) \mid D = 0\right)$$

It asserts that the treated group and the control group would have evolved similarly in the absence of treatment.

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It is **not trivial**, and **requires careful thinking about the outcome variable**. Suppose you are evaluating an active labor market policy. You are interested in estimating the effects on earnings. Do you prefer gross earnings? Log gross earnings? Why does it matter?

The DiD estimator

Sharp treatment

Assume the treatment is **sharp**, meaning that all units in the treated group $D = 1$ get the treatment at period $t = 1$ and nobody in the control group $D = 0$ gets it.

The DiD estimator in this case is :

$$DiD = \mathbb{E}\left(Y_{t=1}(1) - Y_{t=0}(0) \mid D = 1\right) - \mathbb{E}\left(Y_{t=1}(0) - Y_{t=0}(0) \mid D = 0\right)$$

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$$DiD = \underbrace{\mathbb{E}\left(Y_{t=1}(1) - Y_{t=0}(0) \mid D = 1\right)}_{\text{Before-after comparison in treatment group}} - \underbrace{\mathbb{E}\left(Y_{t=1}(0) - Y_{t=0}(0) \mid D = 0\right)}_{\text{Before-after comparison in control group}}$$

It is the difference across groups of the within group differences (name of the method is well-chosen).

The difference-in-differences estimator: an idea

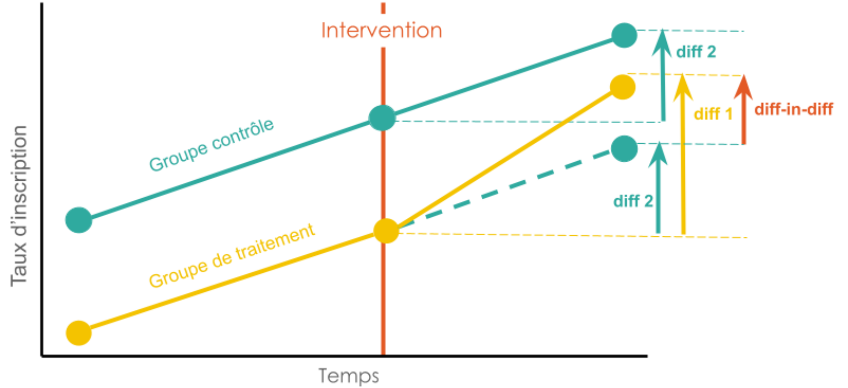

Figure 3: Graphical explanation of the DiD estimator (credit: IDEE program)

The DiD estimates the ATT

Identification of the DiD

Under the parallel trends assumption, sharp treatment and with 2 time periods, the DiD estimator equals the ATT.

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\end{aligned}
$$

The empirical DiD

The usual DiD estimator is :

$$DiD = \mathbb{E}\left(Y_{t=1}(1) - Y_{t=0}(0) \mid D = 1\right) - \mathbb{E}\left(Y_{t=1}(0) - Y_{t=0}(0) \mid D = 0\right)$$

The empirical equivalent to DiD is:

$$\begin{aligned}\widetilde{DiD} =& \frac{1}{N_1}\sum_{i=1}^{N} Y_i \cdot \mathbb{1}(D_i = 1) \cdot \mathbb{1}(t = 1) - Y_i \cdot \mathbb{1}(D_i = 1) \cdot \mathbb{1}(t = 0) \\ &- \left(\frac{1}{N_0}\sum_{i=1}^{N} Y_i \cdot \mathbb{1}(D_i = 0) \cdot \mathbb{1}(t = 1) - Y_i \cdot \mathbb{1}(D_i = 0) \cdot \mathbb{1}(t = 0)\right)\end{aligned}$$

where N_1 is the number of units in the treated group, N_0 is the number of units in the control group.

This estimator is unbiased for the causal effect under the parallel trends assumption.

Which regression can give us $\widetilde{DiD}$ and its standard error?

The DiD estimator and regression

The regression:

$$Y = \underbrace{\alpha_0 + \alpha_1 \cdot \mathbb{1}(D = 1)}_{\text{Group fixed effects}} + \underbrace{\beta_1 \cdot \mathbb{1}(t = 1)}_{\text{Time fixed effects}} + \delta \cdot \mathbb{1}(t = 1) \cdot \mathbb{1}(D = 1)$$

The coefficient δ will be $\widetilde{DiD}$.

Why?

Parallel trends and fixed effects model

The parallel trends assumption is:

$$\mathbb{E}\left(Y_{t=1}(0) \mid D=1\right)-\mathbb{E}\left(Y_{t=0}(0) \mid D=1\right)=\mathbb{E}\left(Y_{t=1}(0) \mid D=0\right)-\mathbb{E}\left(Y_{t=0}(0) \mid D=0\right)$$

which is equivalent to:

$$\mathbb{E}\left(Y_{t=1}(0) \mid D=i\right)= \underbrace{\mathbb{E}\left(Y_{t=0}(0) \mid D=i\right)}_{\text{Group fixed outcome at baseline}} + \underbrace{\mathbb{E}\left(Y_{t=1}(0) \mid D=j\right)-\mathbb{E}\left(Y_{t=0}(0) \mid D=j\right)}_{\text{Common trends from period 0 to 1 across groups}}$$

This means that the untreated potential outcome of any group is the sum of (i) a group specific parameter and (ii) a time trend parameter common to all groups.

$\Rightarrow$ Under the classical DiD framework, the DiD can be estimated by a two-way fixed effect regression.

Revisiting Card & Krueger

Estimating the effect of the minimal wage natural experiment

On Moodle, you have access to the original dataset by Card and Krueger (1994). You can load it using:

```
library(arrow)
read_parquet("Card_Krueger_1994.parquet")
```

The following variables will be useful:

- **empft (Employment Full-Time):** The number of full-time employees (non-managers) at the restaurant.
- **nmgrs (Number of Managers):** The total number of managers working at the restaurant.
- **emppt (Employment Part-Time):** The number of part-time employees at the restaurant.

Build the main FTE employees variable from the Card and Krueger paper, both before and after treatment. Hint: (i) they use the approximation that part-time employees are half-time employees; (ii) variable_name2 indicates value of the variable after treatment.

Estimating the effect of the minimal wage natural experiment

The following code should work:

```
        df <- df%>%
                mutate(fte=empft+nmgrs+(0.5*emppt),
                        fte_after=empft2+nmgrs2+(0.5*emppt2))
```

Compute the average value of these variables in New Jersey and Pennsylvania before and after treatment. Compute the diff-in-diff estimate using these means.

Estimating the effect of the minimal wage natural experiment

The following code should work:

```
PA_diff <- (mean(df[df$state==0,]$fte_after,na.rm=T)-mean(df[df$state==0,]$fte,na.rm=T))
NJ_diff <- (mean(df[df$state!=0,]$fte_after,na.rm=T)-mean(df[df$state!=0,]$fte,na.rm=T))

# the difference in differences
did_mean <- (NJ_diff-PA_diff)
```

You can see that you obtain the same point estimate as Card and Krueger (1994) – the small difference is due to their handling of closed stores.

Estimate the DiD using a linear regression. You can find a version of the dataset fit for this use on the Moodle, it is named "Card_Krueger_Reg.parquet".

Estimating the effect of the minimal wage natural experiment

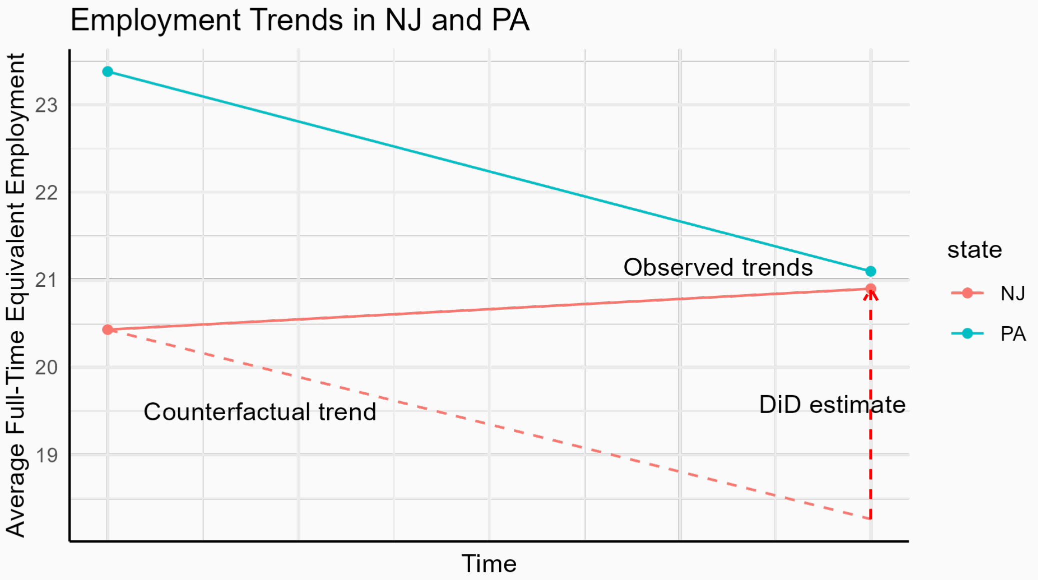
The following code should work:

```
lm <- lm(fte ~ state+post_reform + state*post_reform, data=df)
summary(lm)
```

Note: standard errors will differ because we do not observe closed stores as well as Card and Krueger do – they treat differently permanently and temporarily closed stores. Moreover, standard errors raise a question in the DiD context because we observe the same unit across time. You should thus not use the classical standard errors in this case.

Can you guess why?

Estimating the effect of the minimal wage natural experiment



Extension of the classical DiD

Parallel trends with more than 2 periods

We can extend the parallel trends assumption to the case when there are more than 2 periods, e.g. $t \in (0, 1, 2, 3, \ldots, T)$. In this case, assume $Y(\mathbf{0_t})$ is the outcome for never-treated individuals at period t.

Parallel trends assumption

$$\mathbb{E}\left(Y(\mathbf{0_t}) - Y(\mathbf{0_{t-1}}) \mid D = 1\right) = \mathbb{E}\left(Y(\mathbf{0_t}) - Y(\mathbf{0_{t-1}}) \mid D = 0\right)$$

It asserts that the treated group and the control group would have evolved similarly in the absence of treatment, for any two consecutive periods.

Parallel trends and fixed effects

$$\mathbb{E}\left(Y(\mathbf{0_t}) \mid D=g\right) = \mathbb{E}\left(Y(\mathbf{0_0}) \mid D=g\right) + \sum_{j=1}^{t} \left(\mathbb{E}\left(Y(\mathbf{0_j}) \mid D=g\right) - \mathbb{E}\left(Y(\mathbf{0_{j-1}}) \mid D=g\right)\right)$$

Under the parallel trends assumption, $\mathbb{E}\left(Y(\mathbf{0_j}) \mid D\right) - \mathbb{E}\left(Y(\mathbf{0_{j-1}}) \mid D\right)$ is constant for all groups D, it only depends on the period j. It is thus a time fixed effect γ_j. Similarly, $\mathbb{E}\left(Y(\mathbf{0_0}) \mid D\right)$ is always taken at period 0, so the variation in this parameter is only across groups. It is a group fixed effect α_g.

$\Rightarrow$ Under the DiD framework with multiple periods but only two groups, the DiD can be estimated by a two-way fixed effect regression.

Event studies graphs

Main difference with simple TWFE: add interaction terms. Also assume that in addition to T periods after treatment, you observe L periods before any treatment so that $t \in (-L, -L+1, \ldots, 0, 1, \ldots, T)$.

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Let τ_t be time fixed effects, and α_i unit fixed effects, $t = 0$ be the index of the pre-treatment period (careful: I use 0 for the sake of consistency in this course, but most authors use $t = -1$ for pre-treatment period).

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Let τ_t be time fixed effects, and α_i unit fixed effects, $t = 0$ be the index of the pre-treatment period (careful: I use 0 for the sake of consistency in this course, but most authors use $t = -1$ for pre-treatment period).

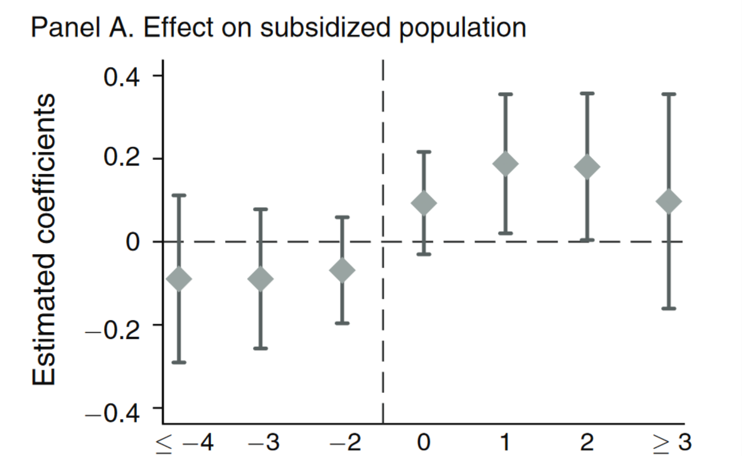
Run the regression :

$$Y_{it} = \sum_{k \neq 0} D_i \times \beta_k \times \mathbb{1}\left(t = k\right) + \alpha_i + \tau_t + \epsilon$$

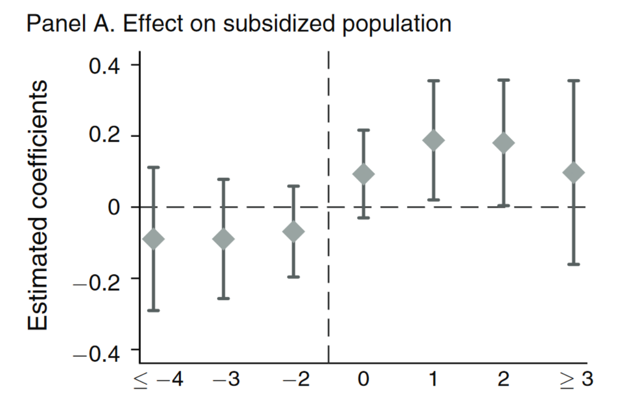
Why is it called event studies ?

Event studies graphs (borrowing from Jonathan Roth)

Graph from He, Guojun, and Shaoda Wang. "Do College Graduates Serving as Village Officials Help Rural China?" American Economic Journal: Applied Economics 9, no. 4 (October 2017): 186–215. https://doi.org/10.1257/app.20160079.



People tend to check parallel trends using these kinds of graphs. What do you think about this practice?



"Pre-test with Caution" (Roth, 2022 – again borrowing his slides)

Main risks:

- Low power for pre-trends test;
- Risk of bias.

Also remind: parallel pre-trends is NOT what we need (so even if they hold, we are still making an assumption about parallel trends after treatment).

Conclusion

Cases when you should be careful and intuitions

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3. When the vector of treatment assignment matters, not just the current treatment. Why should you be cautious?
   $\Rightarrow$ Because in this case, the control group needs to be appropriate, and TWFE and DiD do not necessarily match anymore.

But don't be desperate... solutions exist

Are these all the cases we should be careful about? No, you should be careful whenever the assumptions we saw are not met (sharp treatment, 2 groups, 2 periods, no dynamic effects, no anticipation of future treatment status).

Does this mean we cannot estimate treatment effects in more complicated designs? No, the DiD literature is very active, and you will see extensions of this simple DiD later (including for parallel trends assumption).

To go further

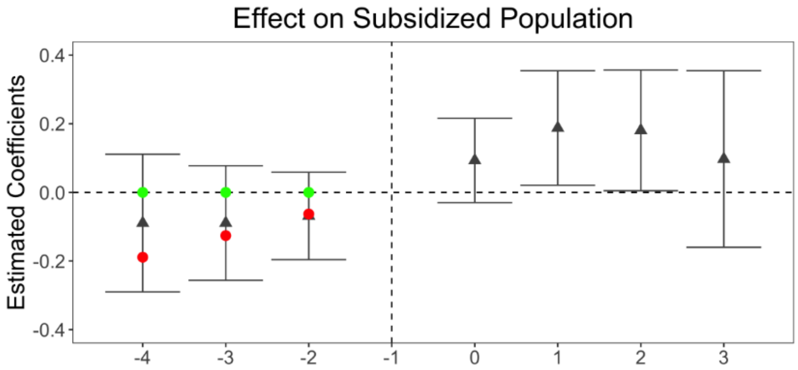
A first step would be to read: Roth, Jonathan, Pedro H. C. Sant'Anna, Alyssa Bilinski, and John Poe. 2023. "What's Trending in Difference-in-Differences? A Synthesis of the Recent Econometrics Literature." Journal of Econometrics 235 (2): 2218–44. https://doi.org/10.1016/j.jeconom.2023.03.008.

Questions?

Presentation by Julia Paul-Venturine

More slides

Parallel pre-trend isn't the only thing in the conf int



- P-value for $H_0 : \beta_{pre} =$ green dots (no pre-trend): 0.81
- P-value for $H_0 : \beta_{pre} =$ red dots: 0.81

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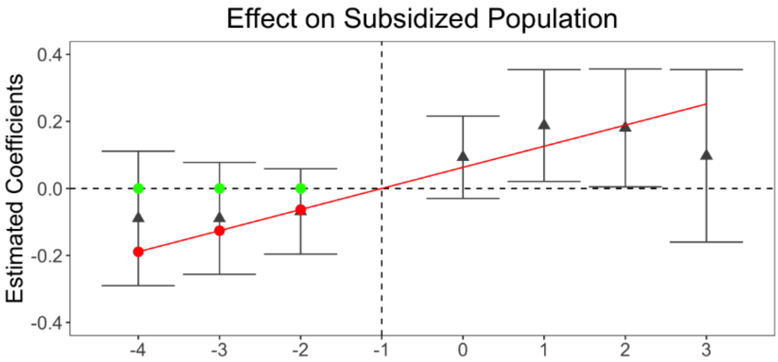

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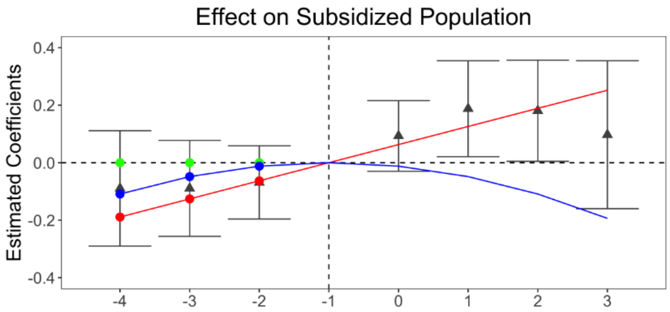
- P-value for $H_0 : \beta_{pre} =$ red dots: 0.81

Parallel pre-trend isn't the only thing in the conf int



- P-value for $H_0 : \beta_{pre} =$ green dots (no pre-trend): 0.81
- P-value for $H_0 : \beta_{pre} =$ red dots: 0.81
- P-value for $H_0 : \beta_{pre} =$ blue dots: 0.81

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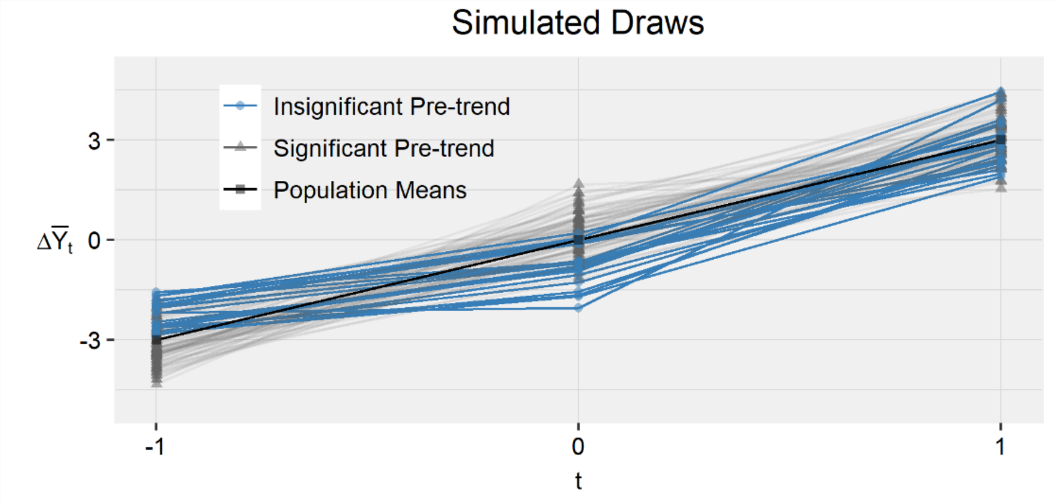
Now, for the "pre-trend bias": some sort of regression to the mean



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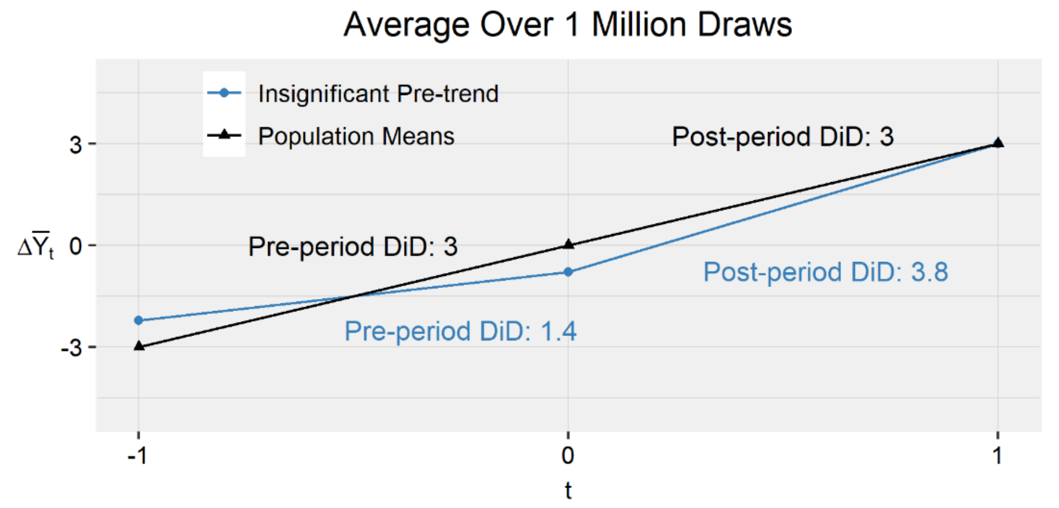

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